ARIMA Modelling of the French Industrial Production Index (IPI)

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0 Introduction

This report focuses on the ARIMA modeling and forecasting of the French Industrial Production Index (IPI) series specifically for the Food Industries sector. The data was obtained from the INSEE's time series databank and contains monthly observations from January 1990 to April 2024. The series is adjusted for seasonal variations and working days (CVS-CJO).

1 The Data

1 Description

The chosen series represents the industrial production index for the food industries sector in France. The index is based on 2021 as the base year (index=100). This series reflects the production output in the food industry, adjusted for seasonal variations and working days.

2 Transformations and Stationarity

The augmented Dickey-Fuller (ADF) test was conducted to determine the stationarity of the series. The ADF test checks for the presence of a unit root in the time series, which would indicate non-stationarity.

The null hypothesis of the ADF test is that the series has a unit root (i.e., it is non-stationary). The alternative hypothesis is that the series is stationary.

Before proceeding with the ADF test, we need to know whether there is a linear trend/intercept in the time series. After performing a simple linear regression we find that both the coefficients for intercept and slope are significant (although this cannot be confirmed since the residuals might be auto-correlated). So we perform an ADF test with constant and trend included. However, an ADF test is only valid in the absence of residual autocorrelation, which we can check for using Ljung-Box tests. We chose the smallest ADF lag for which we never reject the absence of residual autocorrelation up to order 24 (2 years). We find that for this time series, the smallest lag is 8, for which the ADF test gives a p-value of 0.3486, which means we cannot reject the unit root.

Since we could not confirm that the previous series is stationary we performed the same analysis on the differentiated series. Now we find no significant intercept and linear trend so we perform ADF tests without constants and trends. We find that the smallest ADF lag for which the absence of autocorrelation for 24 months is never rejected is lag = 6. In this case, the ADF test gives a p-value of 0.01, so we can reject the unit root and conclude that the differentiated series is stationary.

3 Graphical representation of both series

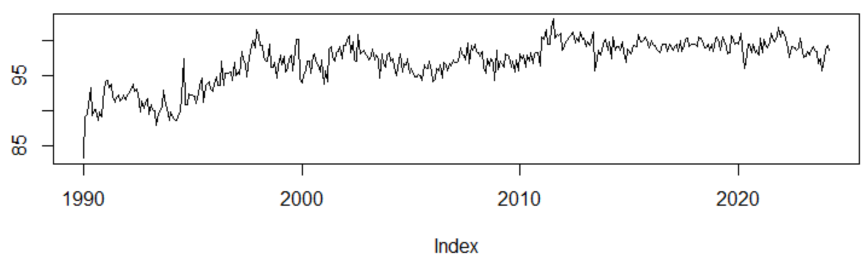

Figure 1: Original Series: French IPI - Food Industries

In the original series, there seems to be an increasing trend, while the differentiated series looks stationary which is consistent with our analysis in the previous section.

2 ARMA Models

4 Model Selection

We will now study the total and partial auto-correlation functions of the differentiated series.

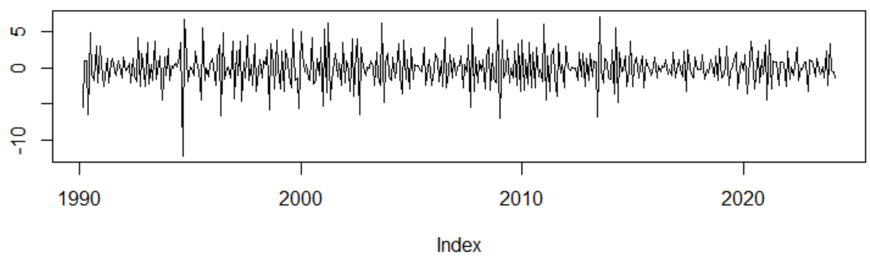

Figure 2: Differenced Series

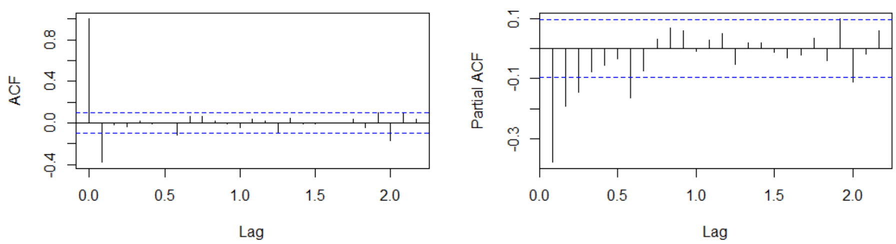

Figure 3: ACF and PACF of differentiated series

We can see in Figure 3, that we should choose a maximum p of 7 (partial autocorrelations no longer significant after lag 7) and a maximum q = 1 for our ARMA models (autocorrelations no longer significant after lag 1). So we evaluated ARMA models for all p less than 7 and q less than 1 and evaluated them based on the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The ARMA(1,1) model was selected as the best model based on AIC, and the ARMA(0,1) model was selected based on BIC.

Model	AIC	BIC
ARMA(1,1)	1410.858	1422.906
ARMA(0,1)	1413.708	1421.74

Table 1: Model Selection based on AIC and BIC

5 Parameter Estimation and Model Validation

We can see from Tables 2 and 3 that the ratio between the estimated coefficients and their standard error is always statistically significant (> 1.96), so both of these models are well-adjusted. We can now perform residual auto-correlation tests (again using the Ljung-Box methodology). In Table 4 we can see that many of the p-values for the ARMA(0,1) model show that there is significant auto-

	Coefficient	**Standard Error**
ar_1	0.1803	0.0783
ma_1	-0.7321	0.0548

Table 2: Coefficients of the ARMA(1,1) model

	Coefficient	**Standard Error**
ma_1	-0.6139	0.0487

Table 3: Coefficients of the MA(1) model

correlation between the residuals, while this is not the case for the ARMA(1,1) model. So we will take the ARMA(1,1) as our final model.

Lag	**p-value (ARMA(1,1))**	**p-value (ARMA(0,1))**
1	NA	NA
2	0.2734	0.0792
3	0.5485	0.1288
4	0.7354	0.2232
5	0.8407	0.2505
6	0.8765	0.2102
7	0.5342	0.0724
8	0.3904	0.0583
9	0.1489	0.0113
10	0.1264	0.0084
11	0.1739	0.0141
12	0.2225	0.0206
13	0.2643	0.0298
14	0.3327	0.0444
15	0.2397	0.0287
16	0.2941	0.0417
17	0.3565	0.0587
18	0.4208	0.0801
19	0.4883	0.1068
20	0.5552	0.1368
21	0.6041	0.1602
22	0.6330	0.1944
23	0.6701	0.2322
24	0.1819	0.0466

Table 4: p-values for ARMA(1,1) and ARMA(0,1) residuals (Ljung-Box test)

This means that our differentiated time series should satisfy:

$$X_t = -0.1803X_{t-1} + \varepsilon_t - 0.7321\varepsilon_{t-1}$$

6 ARIMA Model

Given the differencing required to achieve stationarity, the parameter "d" will equal 1. This means that the original series will be modeled by an

$$\text{ARIMA}(1, 1, 1)$$

The original series Y_t should satisfy:

$$Y_t - 0.8197Y_{t-1} - 0.1803Y_{t-2} = \varepsilon_t - 0.7321\varepsilon_{t-1}$$

since

$$\phi(B)(1 - B)^d Y_t = \psi(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2), \quad d = 1.$$

3 Prediction

7 Confidence Region for Future Values

Denote T as the length of the series. Assume the series' residuals are Gaussian. We know that $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 > 0$. Hence, given X_T, ε_T we have:

$$\begin{cases} X_{T+1|T} = bX_T + c\varepsilon_T = X_{T+1} - \varepsilon_{T+1} \\ X_{T+2|T} = bX_{T+1|T} = bX_{T+1} - b\varepsilon_{T+1} = X_{T+2} - c\varepsilon_{T+1} - \varepsilon_{T+2} - b\varepsilon_{T+1} \end{cases} \tag{1}$$

In our case $b = -0.1803$ and $c = -0.7321$

Let $X' := \begin{pmatrix} X_{T+1|T} \\ X_{T+2|T} \end{pmatrix}$ and $X'' := \begin{pmatrix} X_{T+1} \\ X_{T+2} \end{pmatrix}$ we can rewrite the initial system of equations as: $X'' - X' = \begin{pmatrix} \varepsilon_{T+1} \\ (b+c)\varepsilon_{T+1} + \varepsilon_{T+2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ b+c & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{T+1} \\ \varepsilon_{T+2} \end{pmatrix}$

$$\Rightarrow \Sigma := \text{Var}(X'' - X') = \sigma^2 \begin{pmatrix} 1 & 0 \\ b+c & 1 \end{pmatrix} \begin{pmatrix} 1 & b+c \\ 0 & 1 \end{pmatrix} = \sigma^2 \begin{pmatrix} 1 & b+c \\ b+c & (b+c)^2 + 1 \end{pmatrix}$$

Since we have $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, $X' - X'' \sim \mathcal{N}(0, \Sigma)$.
Finally since $Y := (X' - X'')^T \Sigma (X' - X'') \sim \chi^2_2$,
$\forall \alpha \in (0, 1)$ the confidence region of level α is:
$Z_\alpha := \{X \in \mathbb{R}^2 | Y \leq q_{1-\alpha}^{\chi_2^2}\}$,
If we want to consider the individual 95% confidence intervals (1-d), we just have to consider the respective variances and we get:

$$Z_{1,\alpha} := \left\{ x \in \mathbb{R} \;\middle|\; x \in \left[X_{T+1|T} - 1.96\sqrt{\sigma^2},\ X_{T+1|T} + 1.96\sqrt{\sigma^2} \right] \right\}$$

$$Z_{2,\alpha} := \left\{ x \in \mathbb{R} \;\middle|\; x \in \left[X_{T+1|T} - 1.96\sqrt{\sigma^2\left((b+c)^2 + 1\right)},\ X_{T+1|T} + 1.96\sqrt{\sigma^2\left((b+c)^2 + 1\right)} \right] \right\}$$

8 Hypotheses for Confidence Region

The key hypotheses used to derive this confidence region are the following. The differentiated time series follows an $ARMA(1,1)$, with parameters specified in Section 2. The residuals of the series follow a Gaussian distribution centered in zero and with known variance. In a real model, we would need to verify such assumptions and estimate the variance using the empirical variance. This would mean the confidence region would necessarily be larger.

9 Graphical Representation and Comment

The forecasted values along with the 95% confidence intervals (1-D) for the best AIC model are shown in Figure 4.

The 95% confidence region (chi-squared) for the best AIC model is shown in Figure 5.

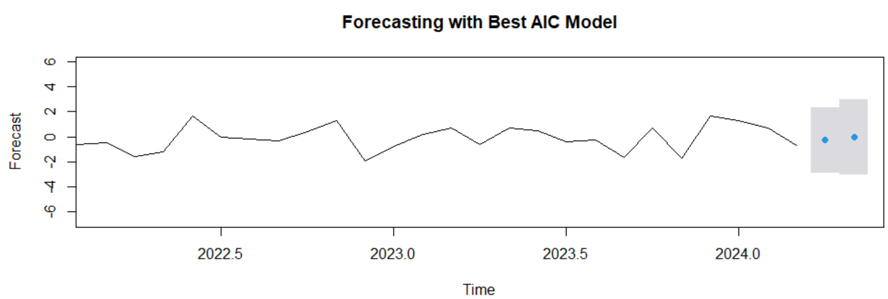

Figure 4: Forecast with Best AIC Model

The plot shows the predicted values with the shaded grey area representing the 95% confidence intervals. The intervals provide a range within which future values are expected to lie with 95% confidence in the next 2 months. The relatively narrow width of the intervals indicates a reasonable level of precision in the forecast, although it may widen as the forecast horizon increases. The blue dots indicate the next 2 predicted values.

10 Open Question

If Y_{T+1} is available faster than X_{T+1}, this information can improve the prediction of X_{T+1} under the condition that Y_t and X_t are correlated. To understand if Y_{T+1} has predictive power over X_{T+1} we can perform a Granger Causality Test. The idea is to compare the following two models:

- **Restricted Model (without Y_t):**

$$X_t = f(X_{t-1}, X_{t-2}, \ldots)$$

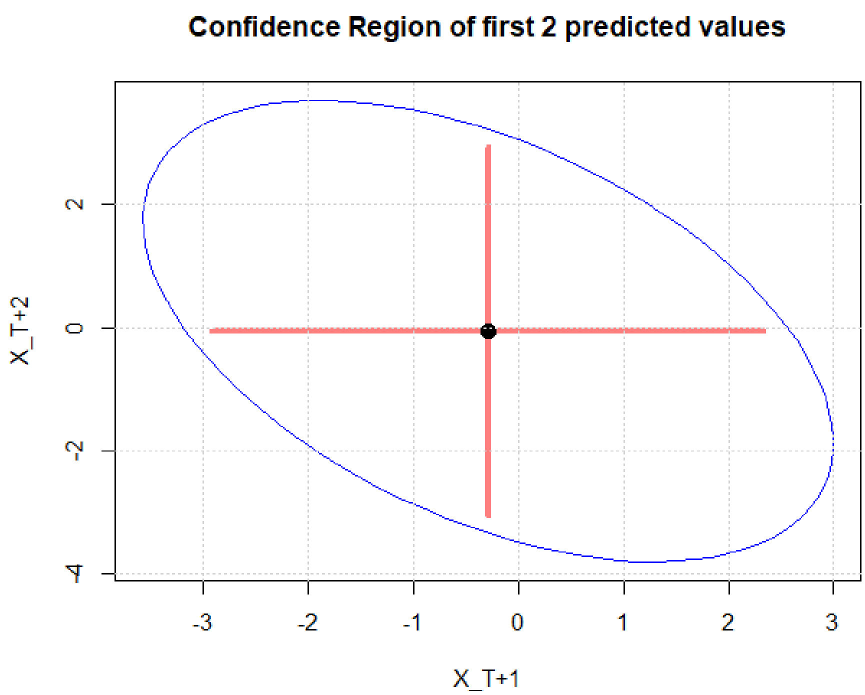

Figure 5: Confidence region (red lines are the 1-dimensional confidence intervals)

- **Unrestricted Model (with Y_t)**:

$$X_t = g(X_{t-1}, X_{t-2}, \ldots, Y_{t-1}, Y_{t-2}, \ldots)$$

The restricted model includes only the past values of X_t, while the unrestricted model includes the past values of both X_t and Y_t. After having estimated both models we can perform an F-Test and decide to either reject the null hypothesis (Y_t does not Granger-cause X_t) or not.

4 Appendix

11 R Code

```
library(readxl)
library(dplyr)
library(zoo)
library(forecast)
library(tseries)
library(fUnitRoots)
library(ellipse)
```

```
rm(list = ls())

# Data preparation
monthly_values <- read_excel("C:\\Users\\mdida\\OneDrive\\Documents\\uni_IPP
     \\Semester␣2\\Time␣Series\\project\\serie_010767602_16052024.xlsx", skip
      = 3)
colnames(monthly_values) <- c("date", "value")
monthly_values$date <- as.yearmon(monthly_values$date, format = "%Y-%m")

date = monthly_values$date
value <- zoo(monthly_values$value, order.by = monthly_values$date)

plot(cbind(value), ylab = "")

# check for linear trends

summary(lm(value ~ date))

# check for autocorrelation in residuals
adf <- adfTest(value, lag=0, type="ct") #ADF test with constant and trend

# autocorrelation tests
Qtests <- function(series, k, fitdf=0) {
  pvals <- apply(matrix(1:k), 1, FUN=function(l) {
    pval <- if (l<=fitdf) NA else Box.test(series, lag=l, type="Ljung-Box",
         fitdf=fitdf)$p.value
    return(c("lag"=l,"pval"=pval))
  })
  return(t(pvals))
}
Qtests(adf@test$lm$residuals, 24,length(adf@test$lm$coefficients))

adfTest_valid <- function(series,kmax,type){ #ADF tests until no more
     autocorrelated residuals
  k <- 0
  noautocorr <- 0
  while (noautocorr==0){
    cat(paste0("ADF␣with␣",k, "␣lags:␣residuals␣OK?␣"))
    adf <- adfTest(series,lags=k,type=type)
    pvals <- Qtests(adf@test$lm$residuals,24,fitdf=length(adf@test$lm$
         coefficients))[,2]
    if (sum(pvals<0.05,na.rm=T) == 0) {
      noautocorr <- 1; cat("OK␣\n")}
    else cat("nope␣\n")
    k <- k + 1
  }
  return(adf)
}
adf <- adfTest_valid(value,24,"ct")
adf


# Difference the series to ensure stationarity
dvalue <- diff(value)
plot(cbind(dvalue), ylab = "")

summary(lm(dvalue ~ date[-1]))
adf <- adfTest_valid(dvalue, 24, type="nc")
adf


# Calculate the ACF
par(mfrow=c(1,2))
acf(dvalue, main = "");pacf(dvalue, main = "")

# Model fitting and storage of results
results <- expand.grid(p = 0:7, q = 0:1)
```

```
results$AIC <- NA
results$BIC <- NA

for (i in 1:nrow(results)) {
  model <- tryCatch({
    arima(dvalue, order = c(results$p[i], 0, results$q[i]), include.mean =
        FALSE)
  }, error = function(e) {
    NULL  # Return NULL if an error occurs
  })

  # Check if model is NULL
  if (!is.null(model)) {
    results$AIC[i] <- AIC(model)
    results$BIC[i] <- BIC(model)
  }
}

# Identify the best model based on AIC and BIC
best_aic_model <- results[which.min(results$AIC),]
print(best_aic_model)
arima101 <- arima(dvalue,c(1,0,1),include.mean=F)
arima101

best_bic_model <- results[which.min(results$BIC),]
print(best_bic_model)
arima001 <- arima(dvalue,c(0,0,1), include.mean = F)
arima001

Qtests(arima101$residuals,24,fitdf=1)
Qtests(arima001$residuals,24,fitdf=1)


# Forecasting with the best model based on AIC
forecasted_values <- forecast(arima101, h = 2, level = 95)

# Determine the range for x-axis to show less of the past values
start_date <- index(dvalue)[length(dvalue) - 24]  # Show the last 24 months
    (2 years) of historical data
end_date <- index(forecasted_values$mean)[2]

# Plot forecast with limited historical data
par(mfrow=c(1,1))
plot(forecasted_values, main = "Forecasting with Best AIC Model", xlab = "
    Time", ylab = "Forecast", xlim = c(start_date, end_date))


# Plot residuals of the best AIC model for diagnostic purposes
par(mfrow = c(2, 1))
ts.plot(residuals(arima101), main = "Residuals of Best AIC Model", ylab = "
    Residuals", xlab = "Time")
acf(ts(residuals(arima101), f = 1), main = "ACF of Residuals of Best AIC 
    Model")


# Extract forecast means and covariance matrix
forecast_means <- forecasted_values$mean[1:2]
residuals_var <- arima101$sigma2
sum_coeffs = sum(arima101$coef)
cov_matrix <- residuals_var * matrix(c(1, sum_coeffs, sum_coeffs, 1 + sum_
    coeffs^2), nrow=2)

# Calculate the 95% confidence ellipse
conf_level <- 0.95
ellipse_points <- ellipse(cov_matrix, centre = forecast_means, level = conf_
    level)

# Plot the forecast and the confidence ellipse
```

```
plot(ellipse_points, type = "n", main = "Confidence␣Region␣of␣first␣2␣
     predicted␣values", xlab = "X_T+1", ylab = "X_T+2")
lines(ellipse_points, col="blue")
segments(forecast_means[1], forecasted_values$lower[2], forecast_means[1],
     forecasted_values$upper[2], col = rgb(1, 0, 0, 0.5), lwd = 4)
segments(forecasted_values$lower[1], forecast_means[2], forecasted_values$
     upper[1], forecast_means[2], col = rgb(1, 0, 0, 0.5),lwd=4)
points(data.frame(t(forecast_means)), cex = 1.5, pch = 19)
grid()
```